phu screen: selecting contigs that carry chosen protein families

A Query models and gene prediction

Query: any mix of

HMM files	Pfam accessions	KOfam orthologs	dbCAN, PULs
capsid.hmm	PF00589	K07497	GH13, PUL0001

Database names are resolved from the pinned phu-db archive into models: **M1** **M2** **M3**

Contigs (FASTA)



→ **pyrodigal / pyrodigal-gv**
meta mode by default; --mode, --ttable

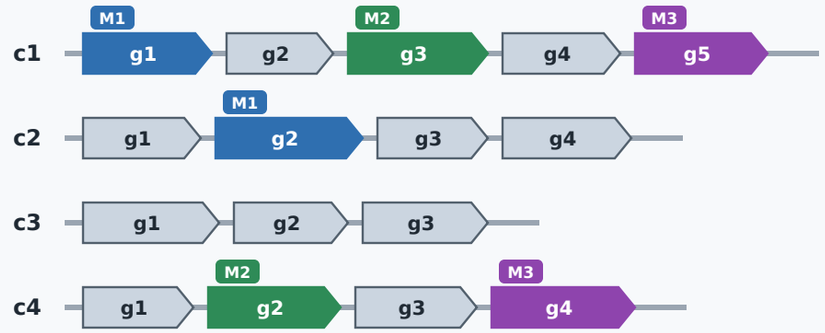
↓

Predicted proteins (≥ 30 aa): **p1** **p2** **p3** **p4** **p5**

Predictions are cached: a new screen of the same contigs with other models skips this step

B Profile search (PyHMMER hmmsearch)

M1 **M2** **M3**



A hit counts when it passes the model's gathering cutoff (--cut-ga, KOfam thresholds) or, for models without one, E-value ≤ 1e-5 (--max-evalue) and --min-bitscore if given. The best hit per model is kept on each contig (--top-per-contig 1).

C Combine hits per contig (--combine-mode)

Which models hit each contig decides whether the contig is kept

	M1	M2	M3	any (default)	all	threshold ≥ 2
c1	●	●	●	kept	kept	kept
c2	●	○	○	kept	dropped	dropped
c3	○	○	○	dropped	dropped	dropped
c4	○	●	●	kept	dropped	kept

any at least one model hits the contig

all every queried model hits the contig

threshold at least --min-hmm-hits models hit the contig

↓

matched contigs

The query and the rule used are written to .phu/query.json

D Optional: turn the hits into new models

Proteins that hit each model, from kept contigs only, can seed a model of your own data

target-proteins/{model}.faa
--save-target-proteins

↓

Align with FAMSA
in-process (pyfamsa); no aligner to install

↓

Trim with trimAl (optional)
--trim-target-msa gappyout | strict | strictplus

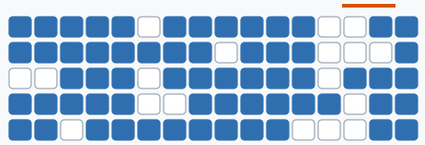
↓

Build the HMM (PyHMMER)
--save-target-hmms

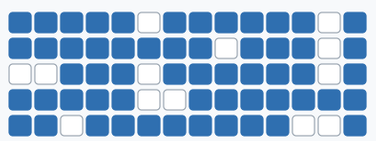
↓

target-hmms/{model}.hmm *reusable as a query in a later phu screen*

alignment gappy



trimmed



Trimming shortens the model and lowers bit scores; off by default.

Outputs

matched-contigs.fna	matched-contig-ids.txt	{hmm}-hmmsearch-domains.tbl	.phu/query.json	target-proteins/, target-hmms/
contigs that passed	their names	per-domain hits, one per HMM file	query, rule, database versions	optional

A matched contig carries sequence similarity to the queried families; it is not a functional or taxonomic assignment.

Contigs, genes and hits shown are illustrative